

[30-1-2026]

[Experiment-1]

EX NO.

Date

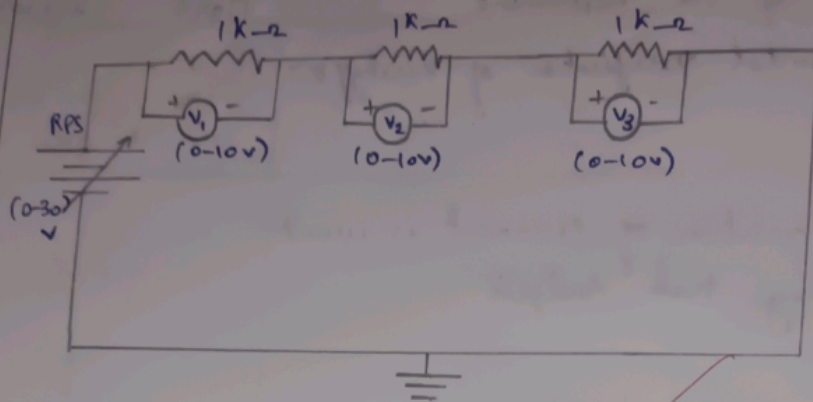
[Experimental Verification of Kirchhoff's Law]Aim :-

To verify Kirchhoff's voltage law and Kirchhoff's current law both theoretically and practically for a given DC circuit.

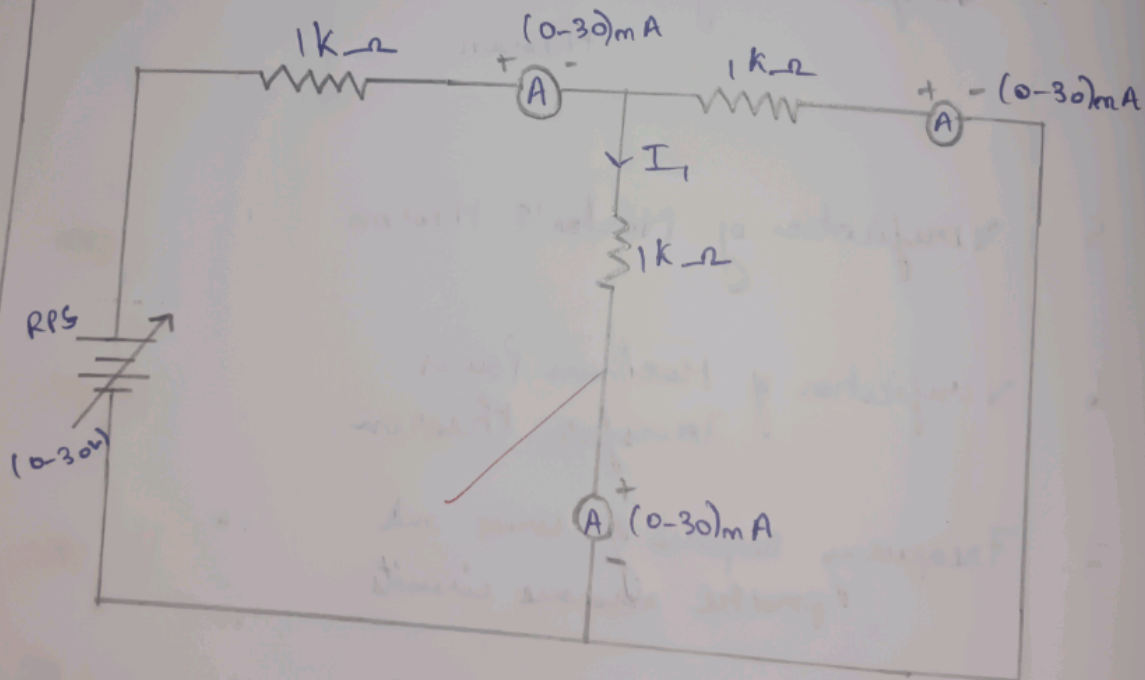
Apparatus Required :-

Sl.No	Apparatus	Specification	Quantity
1.	Regulated Power Supply (RPS)	(0-30)V	1
2.	Voltmeter	(0-30V) MC	3
3.	Ammeter	(0-10mA) MC	3
4.	Resistors	1k Ω	3
5.	Bread board	-	1

[Kirchhoff's Voltage Law]



[Kirchhoff's Current Law]



use
proceed

Tabular column:

V (volts)	V ₁ (Volts)	V ₂ (Volts)	V ₃ (Volts)	V = V ₁ + V ₂ + V ₃ (Volts)	I (amps)	I ₁ (amps)	I ₂ (amps)	I = I ₁ + I ₂ (amps)
2	0.85	0.72	0.72	2 = 0.85 + 0.72 + 0.72	2.09	2.03	1.05	2.08
4	1.43	1.42	1.40	4 = 1.43 + 1.42 + 1.40	2.73	1.38	1.40	2.78
6	1.94	1.89	1.86	6 = 1.94 + 1.89 + 1.86	4.13	2.06	2.05	4.09

Calculations:

For $V = 3V$

KVL:

$$R_{\text{required}} = R_1 + R_2 + R_3 = (1+1+1) = 3\Omega$$

By Ohm's Law,

$$I = \frac{V}{R} = \frac{3}{3 \times 10^3} = 1 \times 10^{-3} A = 1V$$

$$I_1 = I \cdot \frac{R_2}{R_1 + R_2} = 1 \times 10^{-3} \cdot \frac{1 \times 10^3}{2 \times 10^3} = 0.5 \times 10^{-3} A$$

$$V_1 = R_1 I = 1 \times 10^3 \times 1 \times 10^{-3} = 1V$$

$$V_2 = R_2 I = 1 \times 10^3 \times 1 \times 10^{-3} = 1V$$

$$V_3 = R_3 I = 1 \times 10^3 \times 1 \times 10^{-3} = 1V$$

$$I = 1 \times 10^{-3} = 1mA$$

$$V = V_1 + V_2 + V_3$$

$$I = I_1 + I_2 = 2mA$$

$$I_2 = \frac{I \cdot R_1}{R_1 + R_2} = 1mA$$

Theory :-

- Kirchhoff's Current Law (KCL) states that, In an electric network, the algebraic sum of currents meeting at any junction is zero.
- Kirchhoff's Voltage Law (KVL) states that, In an electric network, the algebraic sum of voltages in a closed path is zero.

Procedure :-

- (i) Give connections as per the circuit diagram.
- (ii) Switch on the supply, vary the RPS and set a particular input voltage.
- (iii) Note down the readings of ammeters and voltmeters and calculate them.
- (iv) Vary the RPS for different input voltages and note down the readings for all the meters.
- (v) Reduce the RPS to its minimum value and switch off the supply.
- (vi) Using the calculated values, verify Kirchhoff's laws practically and verify it theoretically.

SL. No	Description	Marks Allocated	Marks obtained
1.)	Experimental setup	20 (marks)	20
2.)	observation / readings	20 (marks)	15
3.)	calculations / result	20 (marks)	18
4.)	Viva	(20 marks)	18
5.)	Record	20 (marks)	18
			92

Result :-

Thus, Kirchhoff's laws are verified theoretically
and practically.

Q.1